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Editors-in-Chief  
Smart Agricultural Technology

Dear Editors,

Please consider the manuscript, “Bounded Instance-Support Evaluation for Finite Crop–Weed Review Queues,” for publication in *Smart Agricultural Technology*.

The study addresses an applied quality-assurance question for agricultural vision: when only a finite number of predicted regions can be reviewed, how much distinct-plant support survives candidate formation, role qualification, ranking, and budget truncation? We organise these stages into an executable evaluation contract for finite crop–weed review queues.

The contract separates spatial support, role-qualified support, merge-aware set coverage, and one-to-one support. The two final endpoints bound geometric plant support when one region creates one quality-assurance record. On the official WeedsGalore split, proposal commissioning nearly doubled spatial support (0.2973 to 0.5713) and raised role-qualified support by 0.1098 to 0.3026. A 27-contract analysis further showed that the queued effect depends on the proposal distribution used to train the ranker.

The model study compares U-Net, DeepLabV3+, and Mask R-CNN under the same  $K = 20$  capacity. It includes crossed training-seed and test-image uncertainty, standard AP and AR, a queue-level PQ-style diagnostic, contract-rank stability, and crop-exposure sensitivity. Mask R-CNN achieved the highest one-to-one point estimate (0.2159); the crossed seed-by-image contrast with U-Net was 0.0132 (95% interval,  $-0.0009$  to  $0.0300$ ).

Two independent operators reviewed the same 518-region queue and reached 0.9421 agreement on weed reviewability (Cohen’s  $\kappa = 0.8930$ ). Merge-aware analysis identified 453 set-covered instances but 348 one-to-one matches, exposing a 105-instance capacity gap that AP, AR, and queue-level PQ-style summaries did not express.

A cross-dataset transfer applied five validation-selected U-Net checkpoints to 60 CWFID images without external fitting. Validation-selected thresholds exceeded the external foreground responses; continuous-score diagnostics located the shift in ranking and threshold response. This experiment connects the finite-queue endpoint to an explicit external-calibration requirement.

The accompanying GitHub release contains figure source data, configurations, per-image and per-seed outputs, matching tables, model and data hashes, preserved run records, and replay programs. The compact verifier reproduces the archived tables and numerical invariants; public-data and model-acquisition instructions support full retraining.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors.

Sincerely,

Zhuo Chen  
on behalf of all authors